. ITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

EXTENDER, TEST POINT, AND RESISTOR, MISCELLANEOUS, KEY CODE FLA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

THIS SPECIFICATION INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic extender, test point, and resistor module.

1.2 Classification. This specification covers one class of modules;

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.
MIL-C-28754	-	Connectors, Electrical, Modular, and Component Parts, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.496 failures per million operating hours when predicted in accordance with MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.4.1 Life test. When tested in accordance with 4.5.3, the resistance shall not exceed ±1.5% of the 25°C post operating temperature dc resistance.

3.4.2 Guide pin receptacle breakout test. The guide pin receptacle shall be capable of withstanding a resultant breakout force of 4.5 pounds (20.02 newtons) applied, when tested in accordance with 4.4.3.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FLA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 226.8 grams maximum.

3.5.4 Contact pins. The module shall have 40 contact pins positioned P1A1 through P1A40.

3.5.5 Depth. The module shall have no additional depth requirements.

3.5.6 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.7 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/125-1, except information normally located on the fin shall be located on the body of the module as shown on figure 1.

3.5.8 Female connector requirement. A female connector J1A1 through J1A40 as specified on figure 2 and equivalent to male connector P1A in electrical contact characteristics shall be provided as part of this module. A grounding bushing shall be part of this connector only if needed to provide electrical contact to exposed metal surfaces. The connector shall be capable of mating with a forty-pin connector identical to P1A.

3.5.9 Datum E on figure 2. All parts of the module below surface E (Datum E) on figure 2 shall conform to the module configuration requirements.

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 3.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

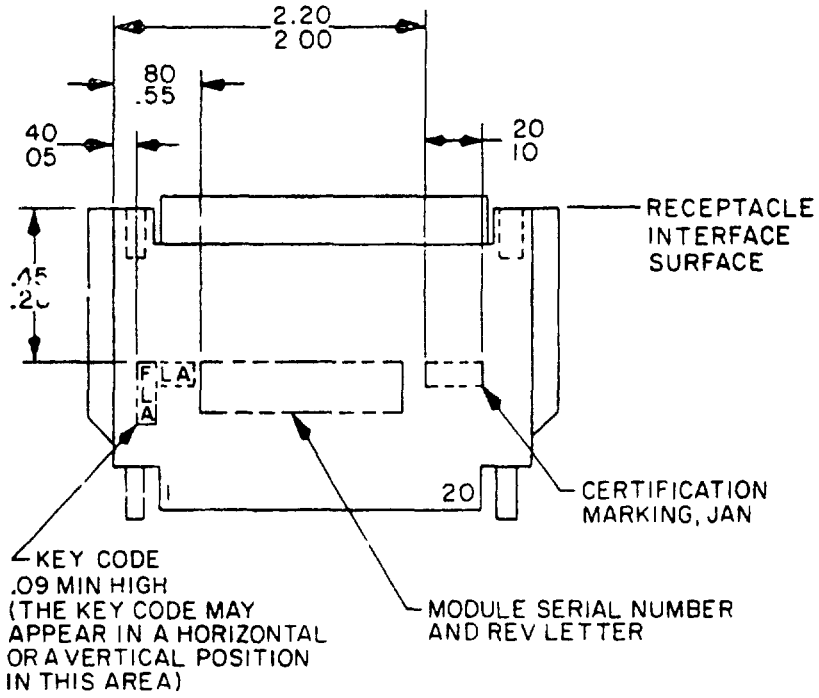
4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures.

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INCHES	MM
.05	1 3
.09	2 3
10	2.5
.20	5.1
.28	7 1
40	10 2
45	11 4
.55	14 0
.80	20.3
2 00	50 8
2 20	55 9

FIGURE 1 Module supplemental marking information

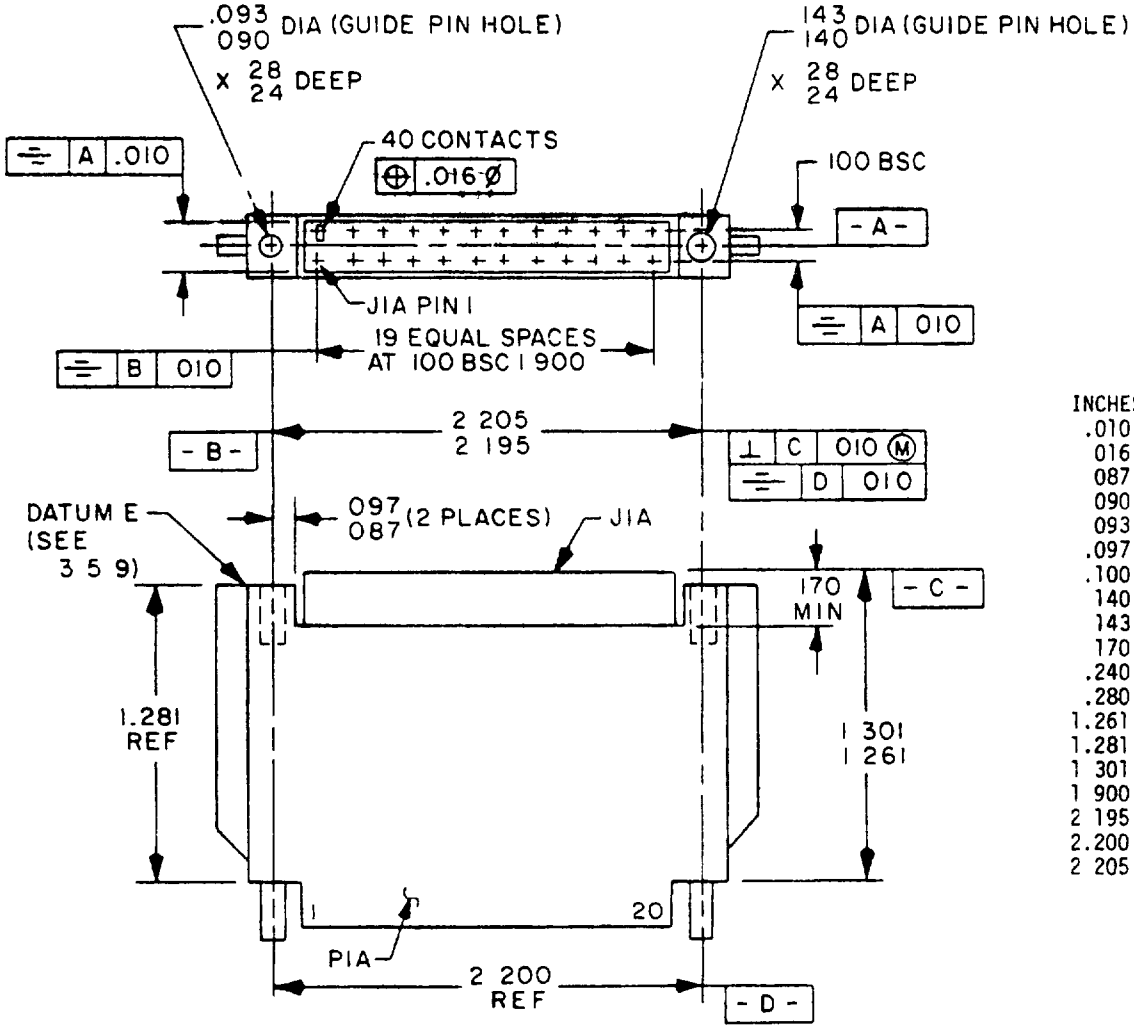
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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/, 2/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Frame isolation	II	X	10	∞	1	∞	---	---	---	---	MΩ
B1-B5	Integrity	III	S	10	∞	1	∞	---	---	---	---	MΩ
C1-C20	DC resistance	IV	X	9.9	10.1	See 3.4.1	See 3.4.1	9.825	10.175	9.678	10.328	kΩ
D1-D20	Continuity	V	S	0	1	0	1	---	---	---	---	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater) at 25°C.

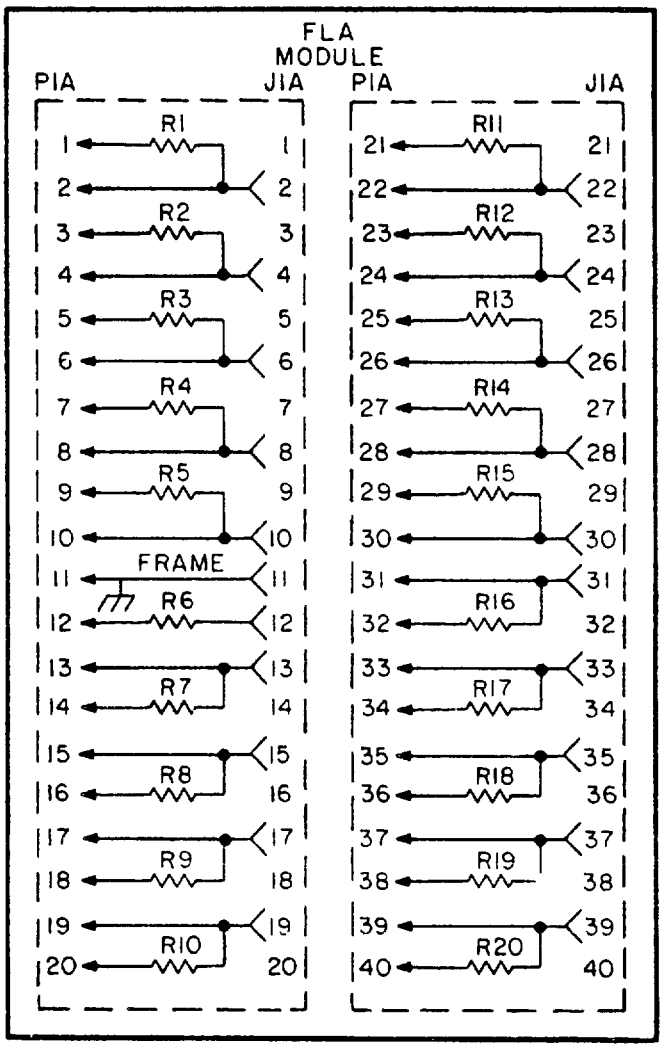
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INCHES	MM
.010	.25
016	41
087	2 21
090	2.29
093	2 36
.097	2.46
.100	2.54
140	3.56
143	3.63
170	4.32
.240	6.10
.280	7 11
1.261	32.03
1.281	32.54
1 301	33.05
1 900	48.26
2 195	55.75
2.200	55.88
2 205	56 01

FIGURE 2 Female connector

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NOTE: R1-R20 = 10 kΩ ±1%, 1/8 W

FIGURE 3. FLA module function diagram, extender, test point, and resistor.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirement of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through D shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows. A, B where applicable, followed by the remaining tests in the sequence.

4.4.3 Guide pin receptacle breakout test. The guide pin receptacle breakout test (Q test) shall be conducted as a part of the mechanical subgroup qualification test. The test shall be conducted in such a manner that when the module is clamped in a test jig it will offer no reinforcement to the guide pin receptacle in any way. Insert a 0.088 ±0.001 inch (2.235 ±0.025 mm) diameter pin and a 0.138 ±0.001 inch (3.505 ± 0.025 mm) diameter pin into their respective guide pin receptacle, making certain the pin seats firmly at the bottom of the receptacle, and apply a force of 4.5 pounds (20.02 newtons) 1 inch (25.4 mm) from the interface surface, in both directions, on each pin along both planes

a. A plane parallel to the connector longitudinal axis and perpendicular to the pin axis.
b. A plane perpendicular to the connector longitudinal axis and perpendicular to the pin axis.

If there is evidence of cracking or degradation, the module shall be rejected for failing this test.

4.4.4 Engaging and separating forces tests. The engaging and separating forces tests (Q test) shall be conducted as a part of the mechanical subgroup qualification test. With the test sample secured to the bed of a testing machine, determine the maximum engaging and maximum separating forces. The maximum engaging force shall be determined using the engaging and separating forces (type III) requirements of MIL-C-28754.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test condition C in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test condition C shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times the module shall be nonoperating.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. During the life test, the module shall be operated as shown on figure 4. Following the life test, but before reducing the temperature of the module, perform test condition C. The resistance values measured shall conform to the 0° to 60°C end-of-life requirements of table I. Following the life test, test condition C shall be performed using the requirements of 3.4.1.

4.5.4 Mechanical tests. In addition to the applicable mechanical tests of MIL-M-28787, the guide pin receptacle breakout test specified in 4.4.3 and the engaging and separating forces tests specified in 4.4.4 shall be performed.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference by classification. Part number M28787/125-1 supersedes part number M28787/125. These two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation		
Resistor	5	mW
Module	100	mW
Input current	1A	maximum

6.5 Module function. The extender, test point, and resistor module when used in conjunction with adapter test point module, 2672173P1, provides up to 20 test points plus chassis frame connection. The module contains twenty 10 kΩ ±1 percent, 1/8 watt resistors which may be pin programmed to provide resistor isolation for the test points. Alternate pin programming may be used to remove resistor isolation for all but one test point. The module may also be used to provide resistive loads but because of the absence of a cooling fin, the total module power dissipation will not exceed 100 mW.

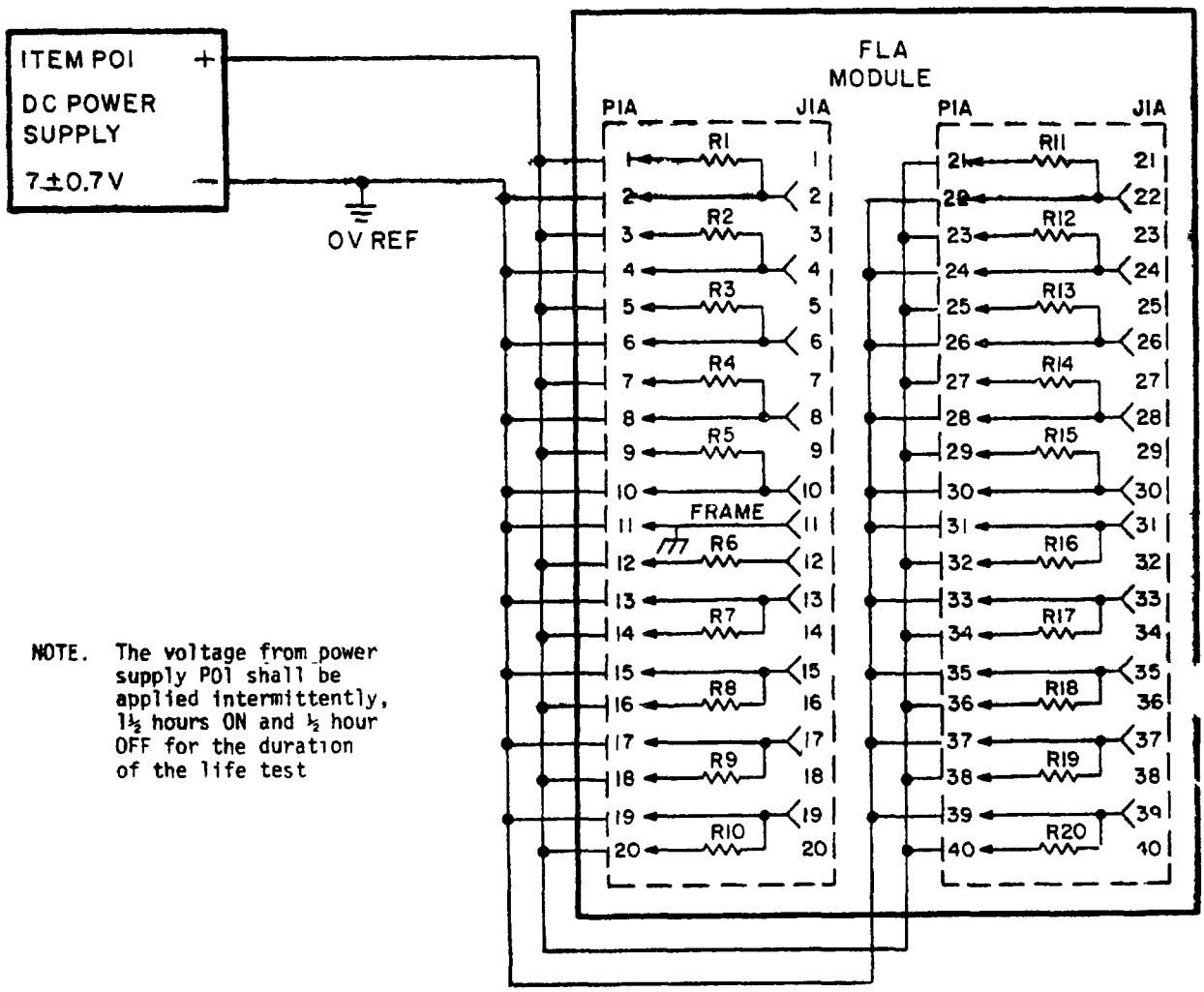
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NOTE. The voltage from power supply P01 shall be applied intermittently, 1½ hours ON and ½ hour OFF for the duration of the life test

FIGURE 4 Life test circuit

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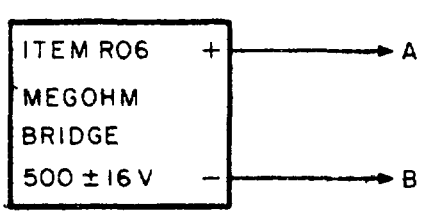
TABLE II. Frame isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Frame isolation	P1All	All other P1A pins	10	∞	MΩ	5

TABLE III. Integrity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to P1A pin	B to P1A pin	Min	Max		
B1	Integrity	1,2,5,6,9,10, 12,15,16,19, 20,21,22,25, 26,29,30,33, 34,37,38	3,4,7,8,11, 13,14,17,18, 23,24,27,28, 31,32,35,36, 39,40	10	∞	MΩ	5
B2		1,2,3,4,9,10, 11,15,16,17, 18,21,22,23, 24,29,30,31, 32,37,38,39,40	5,6,7,8,12, 13,14,19,20, 25,26,27,28, 33,34,35,36,				
B3		1,2,3,4,5,6,7, 8,15,16,17,18, 19,20,21,22, 31,32,33,34, 35,36,37,38	9,10,11,12, 13,14,23,24, 25,26,27,28, 29,30,39,40				
B4		1,2,3,4,5,6, 7,8,9,10,21, 22,23,24,25, 26,27,28,29, 30	11,12,13,14, 15,16,17,18, 19,20,31,32, 33,34,35,36, 37,38,39,40				
B5		1,2,3,4,5,6, 7,8,9,10,11, 12,13,14,15, 16,17,18,19, 20	21,22,23,24, 25,26,27,28, 29,30,31,32, 33,34,35,36, 37,38,39,40				

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WARNING

When using the megohm bridge, high voltage may be present at any of the binding posts. Always set the function switch of the megohm bridge to discharge before connecting or disconnecting the test leads

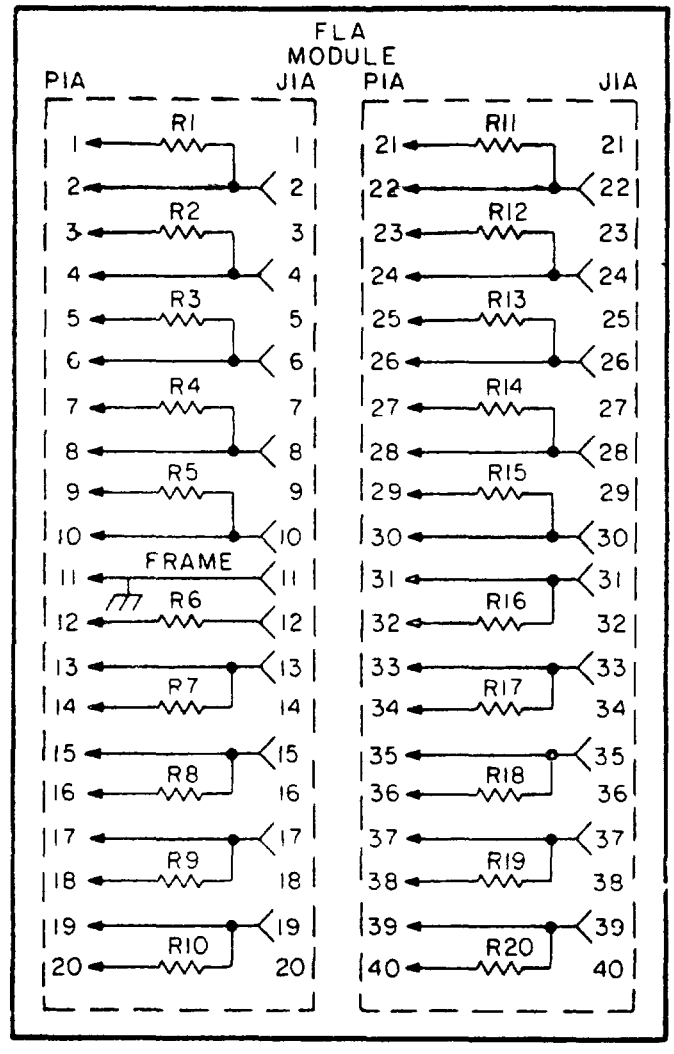


FIGURE 5. Frame isolation and integrity test circuit for tables II and III

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TABLE IV. DC resistance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	DC resistance	P1A1	J1A2	9.9	10.1	kΩ	6
C2		P1A3	J1A4				
C3		P1A5	J1A6				
C4		P1A7	J1A8				
C5		P1A9	J1A10				
C6		P1A12	J1A12				
C7		P1A14	J1A13				
C8		P1A16	J1A15				
C9		P1A18	J1A17				
C10		P1A20	J1A19				
C11		P1A21	J1A22				
C12		P1A23	J1A24				
C13		P1A25	J1A26				
C14		P1A27	J1A28				
C15		P1A29	J1A30				
C16		P1A32	J1A31				
C17		P1A34	J1A33				
C18		P1A36	J1A35				
C19		P1A38	J1A37				
C20		P1A40	J1A39				

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TABLE V. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Continuity	P1A2	J1A2	0	1	Ω	6
D2		P1A4	J1A4				
D3		P1A6	J1A6				
D4		P1A8	J1A8				
D5		P1A10	J1A10				
D6		P1A11	J1A11				
D7		P1A13	J1A13				
D8		P1A15	J1A15				
D9		P1A17	J1A17				
D10		P1A19	J1A19				
D11		P1A22	J1A22				
D12		P1A24	J1A24				
D13		P1A26	J1A26				
D14		P1A28	J1A28				
D15		P1A30	J1A30				
D16		P1A31	J1A31				
D17		P1A33	J1A33				
D18		P1A35	J1A35				
D19		P1A37	J1A37				
D20		P1A39	J1A39				

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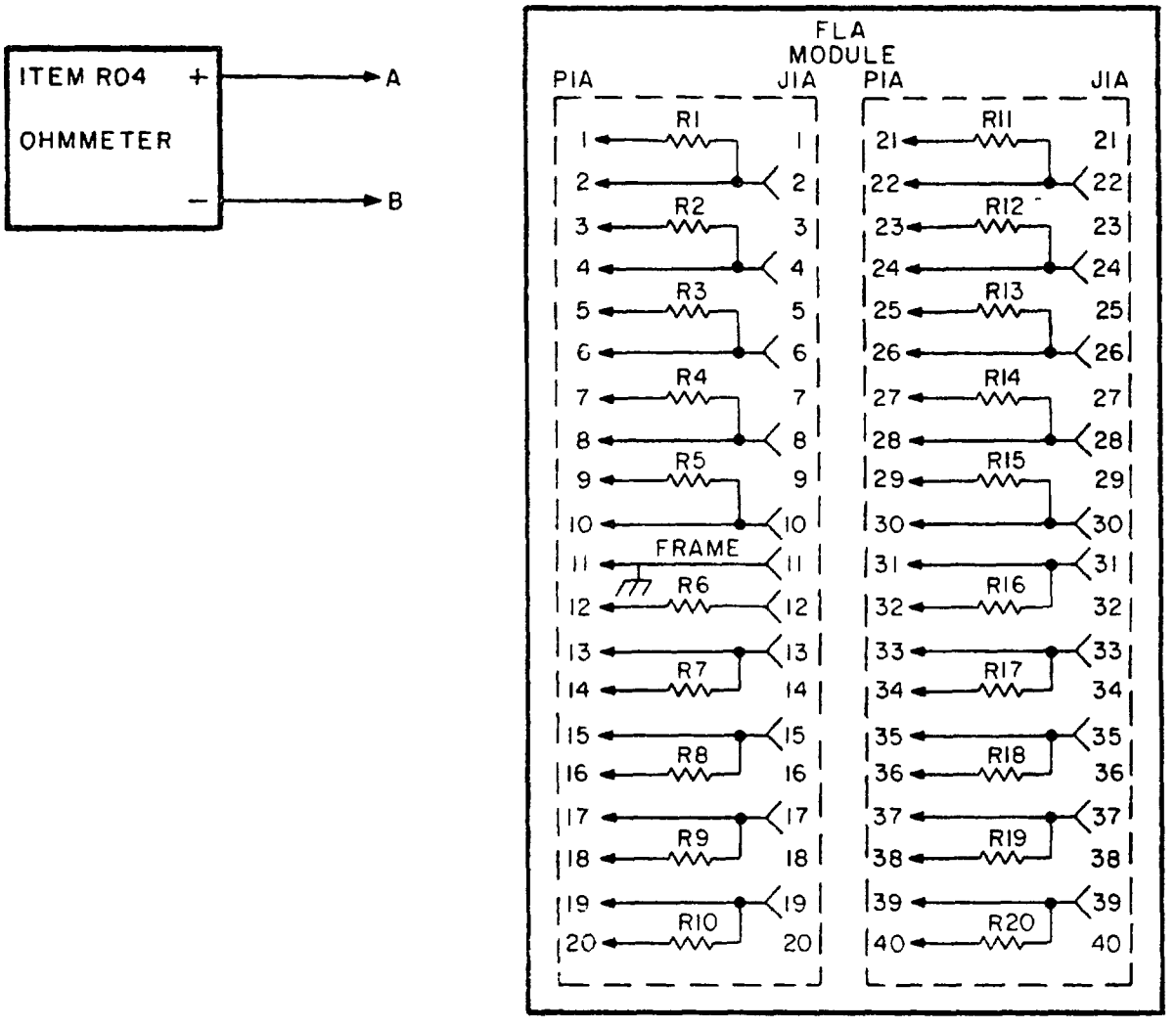


FIGURE 6. DC resistance and continuity test circuit for tables IV and V.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-28787/125A(EC); Modules, Standard Electronic Extender, Test Point, and Resistor, Miscellaneous, Key Code FLA

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